HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and tested embedded C code for resource-constrained microcontrollers, focusing on bare-metal environments.
- Implemented drivers for embedded interfaces including **SPI** and **I²C**, integrating sensor data into the main application flow.
- Debugged embedded software issues using hardware debuggers and JTAG interfaces, resolving critical bugs prior to release.
- Contributed to modular software design discussions and participated in **code reviews** for peer engineers.
- Created and maintained software requirements documents and detailed design specifications in accordance with project timelines.
- Investigated and resolved complex software defects found during integration testing and early hardware prototypes.
- Managed source control using **Git**, including branching strategies and pull request workflows for team collaboration.

SKILLS

Languages – Embedded C, C++, Python

Embedded Systems – Microcontrollers, Bare-metal, RTOS, UART, SPI, I²C, ADC, PWM

Development Tools – JTAG, Hardware Debuggers, Source Control, Git, Jenkins

Documentation – Software Requirements, Design Documentation, Test Records, Risk Analysis

Methodologies – Agile, Scrum, Modular Design, Code Reviews, Unit Testing, Integration Testing

PROJECTS

Real-time Sensor Data Acquisition System

- Developed embedded C firmware for a custom microcontroller board to interface with environmental sensors via **SPI** and **I²C**.
- Implemented a **bare-metal** scheduler to manage concurrent data acquisition from multiple sensors, achieving a sampling rate of 100 Hz.
- Integrated the sensor data processing module with a **UART** interface for transmitting readings to a companion analysis application.

Embedded Python Automation for Test Rigs

- Created **Python** scripts to automate the configuration and validation of embedded hardware test benches.
- Developed modular functions to control embedded interfaces like **PWM** and **ADC**, simulating real-world operating conditions.
- Implemented automated test case execution and generated summary reports, reducing manual testing time by approximately **30%**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)